Supplementary Information
Background matching audits mass spectral antimicrobial resistance prediction

Supplementary Note 1: Why background matching is needed

Raw external AUC answers whether a model ranks resistant isolates above susceptible isolates at a target site. It does not answer whether that ranking is driven by focal-drug biology or by the resistant population in which the label occurs. In bacterial clinical data, resistance labels are structured by lineage, mobile elements, local prescribing ecology and co-resistance. A MALDI-TOF model can exploit this structure because spectra encode abundant cellular proteins and lineage-associated peak patterns.

The background-matched audit controls the observable co-resistance component of this structure. For each focal drug, isolates are grouped by the AST labels of the other drugs in the same panel. The focal drug is excluded from the signature. Within valid strata, resistant and susceptible isolates have the same observed co-resistance background. If the model still separates them, it has residual within-background signal. If performance collapses, the original raw AUC was likely driven partly by background differences between strata.

Supplementary Note 2: Interpretation of the three AUCs

Raw AUC is computed over all available isolates for the focal organism-drug-site comparison. It is the standard external performance metric.

Matched AUC is computed after removing strata that lack enough resistant and susceptible isolates for the focal drug. It asks whether performance remains in the subset where within-background comparisons are possible.

Background-centered AUC subtracts the mean model score within each valid background stratum before pooling and recomputing AUC. It is the strictest metric because it removes score shifts shared by all isolates in a background stratum.

Supplementary Note 3: How to read low-retention rows

Low retention is not a model failure. It is an audit limitation. If a drug-site pair has few strata containing both resistant and susceptible isolates, then the data do not support a strong within-background conclusion. These rows should be reported as cautionary rather than interpreted as biological evidence.

This is why DRIAMS-B ciprofloxacin is not used as main evidence despite a high centered AUC. The estimate is based on only 25 matched isolates and one valid stratum. The method intentionally makes this visible.

Supplementary Figure S1: Cross-resistance network

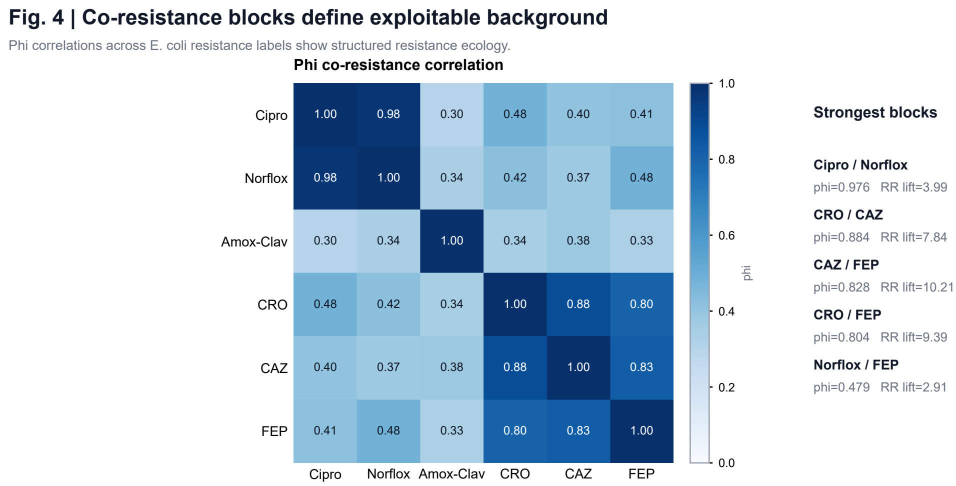

Figure 1: **Cross-resistance network in the expanded *E. coli* panel.** Edge width and shading encode pairwise phi correlation. Two strong blocks are evident: a fluoroquinolone block (ciprofloxacin/norfloxacin, $\phi = 0.976$) and an extended-spectrum cephalosporin block (ceftriaxone, ceftazidime and cefepime, $\phi = 0.804$–0.884). These co-resistance structures provide exploitable background signal for any model predicting drugs within either block. Source data: `source_data_fig4_cross_resistance_edges_all_sites.csv`.

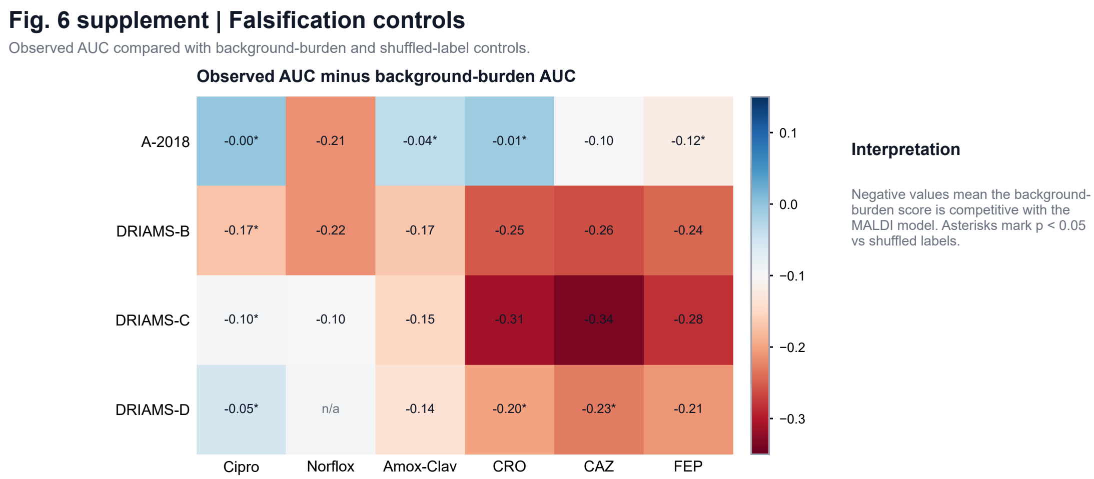

Figure 2: **Falsification controls for background-burden and within-background shuffled-label signal.** Negative observed-minus-burden values indicate rows where a simple non-focal resistance-burden score is competitive with the MALDI model. Shuffled-label controls preserve background strata while disrupting focal-label association.

Table 1: Claim guardrails used throughout the manuscript.

Status	Statement
Supported	MALDI-AMR prediction is background-sensitive, and raw AUC can overstate focal-drug signal.
Supported	Background-centered AUC separates retained within-background ranking from background-driven attenuation.
Supported	Public WGS-linked Bruker data show that ST131 lineage is strongly encoded in MALDI peak features.
Not claimed	The DRIAMS CNN is definitively detecting ST131.
Not claimed	The discriminative DRIAMS or UPEC peaks have definitive protein identities.
Not claimed	All MALDI-AMR performance is clonal confounding.

These guardrails distinguish the evaluation-framework claim from direct clone or protein-identification claims.

Supplementary Figure S2: Falsification controls

Supplementary Figure S3: *S. aureus*/oxacillin audit

Supplementary Figure S4: Three-way decomposition

Supplementary Figure S5: Deployment decision flow

Supplementary Figure S6: Framework comparison

Supplementary Figure S7: Audit atlas summary

Supplementary Table 1: Claim guardrails

Supplementary Table 2: Compact model-family table

Supplementary Table 3: Full *K. pneumoniae* six-drug audit

Supplementary Table 4: Primary background-matched audit

Supplementary Table 5: Full model-family replication

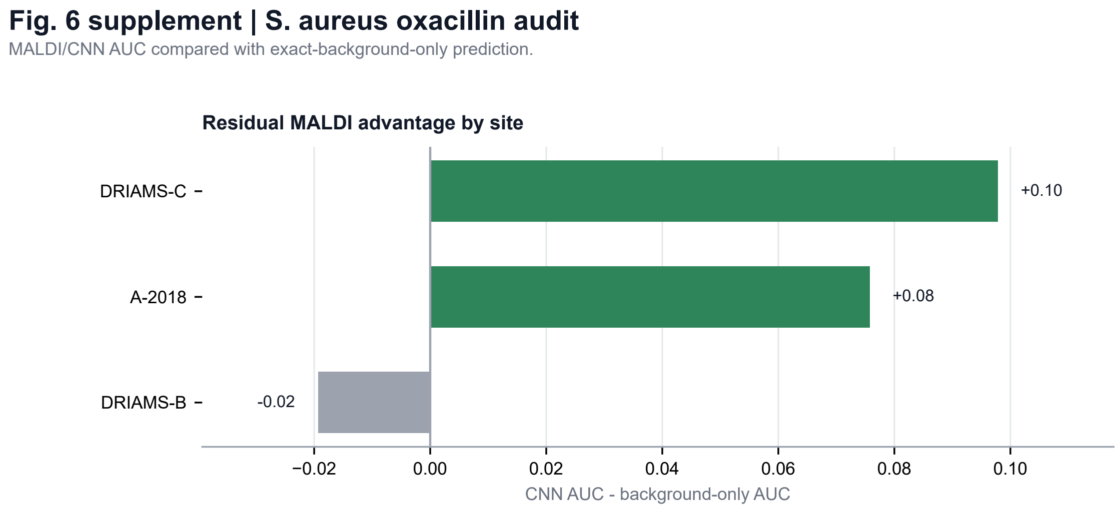

Figure 3: ***S. aureus*/oxacillin second-organism audit.** Oxacillin predictions retained background-centered signal at A-2018 and DRIAMS-C and exceeded the exact-background no-spectrum baseline at both sites. DRIAMS-B is shown for completeness but has only one valid matched stratum. This analysis demonstrates that the audit applies beyond *E. coli*; it is not a systematic multi-organism atlas.

additions would be:

- WGS or MLST labels linked directly to DRIAMS isolates.
- Clone-controlled public WGS-linked MALDI analyses within ST131 and non-ST131 strata.
- Independent MALDI-AMR deployment validation on a non-DRIAMS dataset.
- MS/MS, MALDI-TOF/TOF or LC-MS/MS validation of discriminative m/z peaks.

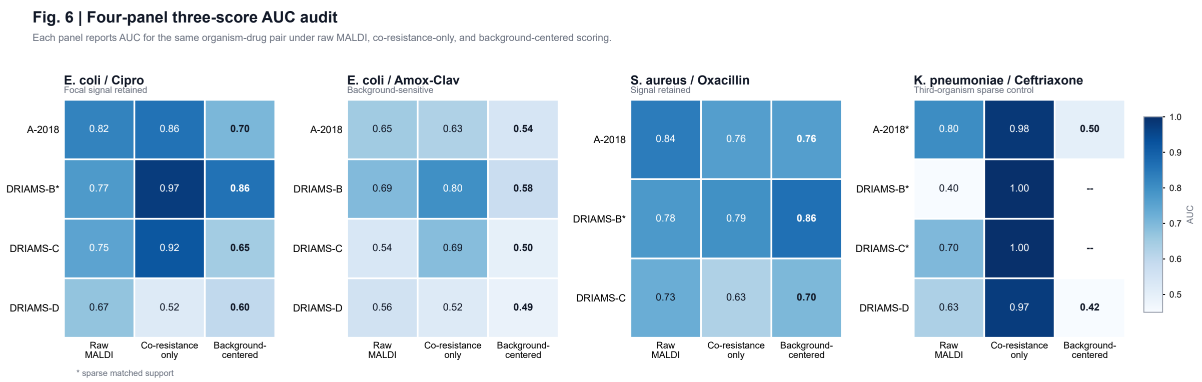

Figure 4: **Four-panel three-score decomposition of MALDI and AST-background signal.** Each panel reports raw MALDI AUC, no-spectrum co-resistance-only AUC and background-centered MALDI AUC for the same organism-drug-site rows. The current version includes the *K. pneumoniae*/ceftriaxone panel as a third-organism sparse-control case.

Table 2: Compact model-family replication of raw-to-background-centered attenuation.

Site	Drug	CNN/Mega	LGBM multi	LGBM single	Weis LR
A-2018	Cipro	0.823-¿0.703	0.828-¿0.640	0.828-¿0.652	0.734-¿0.689
DRIAMS-C	Cipro	0.750-¿0.646	0.814-¿0.637	0.804-¿0.682	0.625-¿0.816*
DRIAMS-D	Cipro	0.671-¿0.596	0.723-¿0.576	0.712-¿0.591	0.667-¿0.620
A-2018	Amox-Clav	0.650-¿0.541	0.635-¿0.482	0.614-¿0.465	0.564-¿0.529
DRIAMS-C	Amox-Clav	0.535-¿0.497	0.593-¿0.518	0.559-¿0.533	0.626-¿0.578
DRIAMS-D	Amox-Clav	0.557-¿0.486	0.580-¿0.449	0.559-¿0.487	0.652-¿0.524

Cells show raw AUC -¿ background-centered AUC. Asterisks mark caution rows with low matched support. Weis LR cells combine the temporal A-2018 export with existing Weis-style DRIAMS-C/DRIAMS-D compatibility rows; they are compatibility evidence rather than pooled benchmarks with the CNN/LightGBM panel.

Fig. 7 supplement | Deployment decision flow

Audit outcomes translated into conservative validation and deployment actions.

Audit pattern	Decision category	Recommended action
Raw high + background-centered high + acceptable calibration	candidate for controlled deployment	Proceed only with local calibration check, locked thresholds, and ongoing drift monitoring.
Raw high + background-centered high + poor calibration	ranking only recalibrate before clinical use	Use only as a ranking signal until calibration, thresholds, and clinical operating points are locally reset.
Raw high + background-centered low	background dependent retrain locally	Do not deploy as portable focal-drug model; validate locally and retrain or recalibrate on current ecology.
Raw high + underpowered matching	insufficient matched evidence	Collect more matched support or additional external labels before making a deployment claim.
Raw low + background-centered low	not deployment ready	Do not deploy; use for research triage or redesign the model/evaluation target.

Figure 5: **Deployment decision flow.** Audit outcomes map to conservative validation, recalibration, retraining or no-deployment actions depending on retained signal, calibration and matched-support adequacy.

Fig. 7 | Framework comparison

What the background-matched audit adds relative to prior MALDI-AMR publications and TRIPOD+AI.

Evaluation dimension	Prior MALDI-AMR	TRIPOD+AI	This framework
Reports external-site AUC	Yes	Required	Yes
Controls for co-resistance background	No	Not required	Yes
Quantifies signal collapse vs. retention	No	No	Yes
Site-specific evaluation	Yes	Recommended	Yes
Multi-drug panel (> 2 pairs)	Some	N/A	Yes
Second organism validation	No	N/A	Yes
Sensitivity analysis (stratum threshold)	No	No	Yes
Model-agnostic (accepts any CSV)	No	N/A	Yes
Open code + reproducible example data	Partial	N/A	Yes

Prior MALDI-AMR: Weis et al. 2022 (Nat. Med.) and comparable publications. TRIPOD+AI: transparent reporting guideline (2024).

Figure 6: **Framework comparison.** The background-matched audit is positioned relative to prior MALDI-AMR evaluation and general clinical prediction reporting frameworks.

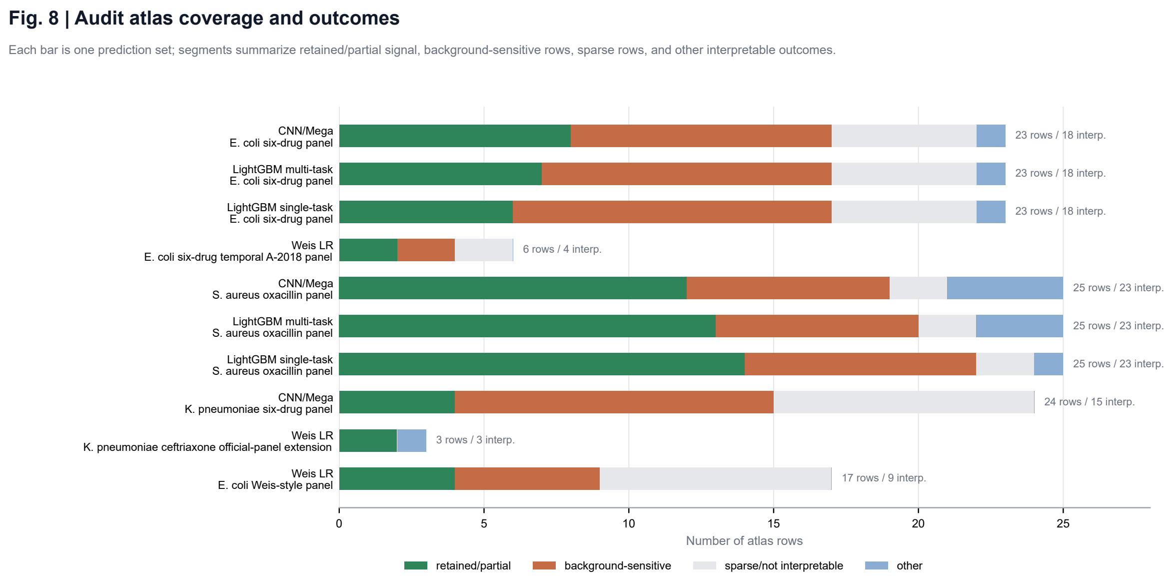

Figure 7: **Audit atlas coverage and outcomes.** Each bar summarizes one prediction set in the current atlas. Segments show retained or partial residual signal, background-sensitive rows, sparse or non-interpretable rows and other interpretable outcomes.

Table 3: Full K. pneumoniae six-drug audit results and Weis ceftriaxone extension.

Model	Site	Drug	Raw AUC	Centered AUC	Background AUC	Retention (%)	Strata	Interpretation
CNN/Mega	A-2018	Amox-Clav	0.743	0.532	0.885	79.6	3	background only competitive and weak residual
CNN/Mega	DRIAMS-B	Amox-Clav	0.331	–	0.994	0.0	0	not interpretable or sparse
CNN/Mega	DRIAMS-C	Amox-Clav	0.646	0.473	0.931	38.6	3	background only competitive and weak residual
CNN/Mega	DRIAMS-D	Amox-Clav	0.612	0.484	0.888	93.2	6	background only competitive and weak residual
CNN/Mega	A-2018	FEP	0.896	–	0.999	0.0	0	not interpretable or sparse
CNN/Mega	DRIAMS-B	FEP	0.443	–	1.000	0.0	0	not interpretable or sparse
CNN/Mega	DRIAMS-C	FEP	0.649	–	1.000	0.0	0	not interpretable or sparse
CNN/Mega	DRIAMS-D	FEP	0.647	0.396	0.998	0.9	1	background only competitive and weak residual
CNN/Mega	A-2018	CAZ	0.809	0.208	0.996	2.1	1	background only competitive and weak residual
CNN/Mega	DRIAMS-B	CAZ	0.371	–	1.000	0.0	0	not interpretable or sparse
CNN/Mega	DRIAMS-C	CAZ	0.679	–	0.999	0.0	0	not interpretable or sparse
CNN/Mega	DRIAMS-D	CAZ	0.623	0.929	0.999	0.5	1	cautionary background only competitive but residual signal
CNN/Mega	A-2018	CRO	0.804	0.500	0.981	3.3	1	background only competitive and weak residual
CNN/Mega	DRIAMS-B	CRO	0.404	–	1.000	0.0	0	not interpretable or sparse
CNN/Mega	DRIAMS-C	CRO	0.698	–	1.000	0.0	0	not interpretable or sparse
CNN/Mega	DRIAMS-D	CRO	0.630	0.424	0.973	81.8	2	background only competitive and weak residual
CNN/Mega	A-2018	Cipro	0.788	0.634	0.796	84.6	5	background only competitive but residual signal
CNN/Mega	DRIAMS-B	Cipro	0.578	0.763	0.809	79.1	1	background only competitive but residual signal
CNN/Mega	DRIAMS-C	Cipro	0.702	0.634	0.939	53.7	3	background only competitive but residual signal
CNN/Mega	DRIAMS-D	Cipro	0.587	0.498	0.598	96.4	6	background only competitive and weak residual
CNN/Mega	A-2018	Piperacillin-Tazobactam	0.738	0.531	0.985	7.7	2	background only competitive and weak residual
CNN/Mega	DRIAMS-B	Piperacillin-Tazobactam	0.376	–	0.923	0.0	0	not interpretable or sparse
CNN/Mega	DRIAMS-C	Piperacillin-Tazobactam	0.665	0.477	0.969	15.1	1	background only competitive and weak residual
CNN/Mega	DRIAMS-D	Piperacillin-Tazobactam	0.598	0.526	0.659	95.3	7	background only competitive and weak residual
Weis LR	DRIAMS-B	CRO	0.509	0.509	–	100.0	1	near chance or ambiguous
Weis LR	DRIAMS-C	CRO	0.672	0.672	–	100.0	1	cautionary moderate within background signal
Weis LR	DRIAMS-D	CRO	0.596	0.596	–	100.0	1	weak or partial within background signal

The CNN/Mega rows summarize the K. pneumoniae six-drug panel. The Weis LR ceftriaxone rows are reported separately because they come from the official-panel compatibility export; background-only AUC was not computed for those rows.

Table 4: Primary background-matched audit for the E. coli ciprofloxacin and amoxicillin-clavulanic acid contrast.

Pair	Site	Raw AUC	Centered AUC	Retention (%)	n matched	P perm	Adequacy	Interpretation
E. coli / Cipro	A-2018	0.823 (0.799-0.849)	0.703 (0.664-0.747)	61.9	836	0.002	interpretable	Retained residual signal
E. coli / Cipro	DRIAMS-B	0.774 (0.692-0.841)	0.860 (0.636-1.000)	11.8	25	0.010	caution_low_n_matched	Caution; low matched support
E. coli / Cipro	DRIAMS-C	0.750 (0.702-0.795)	0.646 (0.576-0.719)	47.6	422	0.002	interpretable	Retained residual signal
E. coli / Cipro	DRIAMS-D	0.671 (0.640-0.700)	0.596 (0.565-0.624)	98.4	1908	0.002	interpretable	Retained residual signal
E. coli / Amox-Clav	A-2018	0.650 (0.619-0.685)	0.541 (0.504-0.576)	92.2	1261	0.038	interpretable	Weak or uncertain residual signal
E. coli / Amox-Clav	DRIAMS-B	0.694 (0.611-0.765)	0.576 (0.410-0.715)	62.1	123	0.204	interpretable	Weak or uncertain residual signal
E. coli / Amox-Clav	DRIAMS-C	0.535 (0.494-0.577)	0.497 (0.449-0.547)	89.2	805	0.675	interpretable	Near chance after matching
E. coli / Amox-Clav	DRIAMS-D	0.557 (0.521-0.590)	0.486 (0.450-0.521)	97.4	1943	0.854	interpretable	Near chance after matching

Centered AUC is computed after subtracting the mean model score within each co-resistance background stratum. Caution rows are reported but not used as the main evidence.

Table 5: Raw and background-centered AUCs across model families for the primary E. coli contrast.

Site	Drug	CNN/Mega raw	CNN/Mega centered	LGBM multi raw	LGBM multi centered	LGBM single raw	LGBM single centered	Weis LR raw	Weis LR centered	Consensus
A-2018	Cipro	0.823	0.703	0.828	0.640	0.828	0.652	0.734	0.689	retained
DRIAMS-C	Cipro	0.750	0.646	0.814	0.637	0.804	0.682	0.625	0.816*	retained
DRIAMS-D	Cipro	0.671	0.596	0.723	0.576	0.712	0.591	0.667	0.620	partial
A-2018	Amox-Clav	0.650	0.541	0.635	0.482	0.614	0.465	0.564	0.529	weak
DRIAMS-C	Amox-Clav	0.535	0.497	0.593	0.518	0.559	0.533	0.626	0.578	collapsed
DRIAMS-D	Amox-Clav	0.557	0.486	0.580	0.449	0.559	0.487	0.652	0.524	collapsed

Columns separate raw and background-centered AUCs for direct model-family comparison. Asterisks mark caution rows with low matched support. Weis LR cells combine the temporal A-2018 export with existing Weis-style DRIAMS-C/DRIAMS-D compatibility rows; they are compatibility evidence rather than pooled benchmarks with the CNN/LightGBM panel.

Table 6: Alternative split-column model-family replication for the primary E. coli contrast.

Site	Drug	CNN/Mega raw	CNN/Mega centered	LGBM multi raw	LGBM multi centered	LGBM single raw	LGBM single centered	Weis LR raw	Weis LR centered	Consensus
A-2018	Cipro	0.823	0.703	0.828	0.640	0.828	0.652	0.734	0.689	retained
DRIAMS-C	Cipro	0.750	0.646	0.814	0.637	0.804	0.682	0.625	0.816*	retained
DRIAMS-D	Cipro	0.671	0.596	0.723	0.576	0.712	0.591	0.667	0.620	partial
A-2018	Amox-Clav	0.650	0.541	0.635	0.482	0.614	0.465	0.564	0.529	weak
DRIAMS-C	Amox-Clav	0.535	0.497	0.593	0.518	0.559	0.533	0.626	0.578	collapsed
DRIAMS-D	Amox-Clav	0.557	0.486	0.580	0.449	0.559	0.487	0.652	0.524	collapsed

This alternate layout separates raw and background-centered AUC columns for direct model-family comparison. Asterisks mark caution rows with low matched support.

Table 7: Strongest co-resistance edges in the E. coli expanded panel.

Drug A	Drug B	n both	n RR	RR lift	Phi	R Jaccard
Cipro	Norflox	1090	263	3.993	0.976	0.963
CRO	CAZ	4349	446	7.841	0.884	0.811
CAZ	FEP	3907	271	10.212	0.828	0.719
CRO	FEP	3956	283	9.394	0.804	0.680
Norflox	FEP	771	89	2.908	0.479	0.392
Cipro	CRO	4371	437	3.223	0.479	0.382

Edges are computed across isolate-level AST labels. RR lift compares observed double resistance with the expectation under independence.

Table 8: Model-class audit comparison with Weis LR compatibility rows.

Site	Drug	CNN/Mega	LGBM multi	LGBM single	Weis LR
A-2018	Cipro	0.823-¿0.703	0.828-¿0.640	0.828-¿0.652	0.734-¿0.689
DRIAMS-C	Cipro	0.750-¿0.646	0.814-¿0.637	0.804-¿0.682	0.625-¿0.816*
DRIAMS-D	Cipro	0.671-¿0.596	0.723-¿0.576	0.712-¿0.591	0.667-¿0.620
A-2018	Amox-Clav	0.650-¿0.541	0.635-¿0.482	0.614-¿0.465	0.564-¿0.529
DRIAMS-C	Amox-Clav	0.535-¿0.497	0.593-¿0.518	0.559-¿0.533	0.626-¿0.578
DRIAMS-D	Amox-Clav	0.557-¿0.486	0.580-¿0.449	0.559-¿0.487	0.652-¿0.524

Cells show raw AUC -¿ background-centered AUC. Asterisks mark caution rows with low matched support. Weis LR cells combine the temporal A-2018 export with existing Weis-style DRIAMS-C/DRIAMS-D compatibility rows; they are not pooled benchmarks with the CNN/LightGBM panel.

Table 9: Public WGS-linked Basel UPEC MALDI validation.

Target	n	positive	AUC	folds	Model
ST131	407	55	0.932	5	centroid_direction
Ciprofloxacin R	350	53	0.755	5	centroid_direction
Ceftriaxone R	360	30	0.689	5	centroid_direction

A simple centroid-direction classifier is used to ask whether lineage and resistance labels are encoded in Bruker MALDI peak features.

Table 10: Mass-matched enrichment of published ST131 MALDI biomarkers among discriminative public UPEC peaks.

Target	Observed	Top peaks	Null mean	Fold enrichment	P perm
ST131	14	75	4.51	3.11	0.0001
Ciprofloxacin R	10	75	4.47	2.24	0.0068
Ceftriaxone R	12	75	4.54	2.64	0.0006
ALL TARGETS	36	225	13.55	2.66	0.0001

The null model preserves the coarse m/z-stratum distribution of selected peaks.

Table 11: WGS-lineage-controlled MALDI prediction in the public Basel UPEC cohort.

Target	Control or subset	n	positive	AUC	Interpretation
ST131	overall	407	55	0.932	lineage is strongly encoded
Ciprofloxacin R	overall	350	53	0.755	unadjusted resistance signal
Ciprofloxacin R	ST131 only	47	30	0.725	residual signal within ST131
Ciprofloxacin R	non-ST131 only	303	23	0.644	weaker residual signal outside ST131
Ciprofloxacin R	exact-ST centered	125	–	0.564	attenuated after sequence-type control
Ceftriaxone R	overall	360	30	0.689	unadjusted resistance signal
Ceftriaxone R	ST131 only	48	21	0.515	near chance inside ST131
Ceftriaxone R	exact-ST centered	147	–	0.503	chance after sequence-type control

Exact-ST centered rows report retained isolates from sequence types containing both resistant and susceptible isolates for the focal drug. These analyses use WGS metadata from the public UPEC cohort and are independent of the DRIAMS-trained CNN.

Table 12: What the background-matched audit adds relative to prior work and TRIPOD+AI.

Dimension	Prior MALDI-AMR	TRIPOD+AI	This framework
Reports external-site AUC	Yes	Required	Yes
Controls for co-resistance background	No	Not required	Yes
Quantifies signal collapse vs. retention	No	No	Yes
Site-specific evaluation	Yes	Recommended	Yes
Multi-drug panel (¿ 2 pairs)	Some	N/A	Yes
Second organism validation	No	N/A	Yes
Sensitivity analysis (stratum threshold)	No	No	Yes
Model-agnostic (accepts any CSV)	No	N/A	Yes
Open code + reproducible example data	Partial	N/A	Yes

Prior MALDI-AMR refers to Weis et al.\{} 2022 (\{}textit{Nature Medicine}) and comparable publications. TRIPOD+AI refers to the transparent reporting guideline (2024). N/A denotes dimensions not applicable to a reporting standard.

Table 13: Audit atlas coverage and outcome summary.

Atlas set	Model	Scope	Rows	Drugs	Sites	Interpretable	Retained/partial	Background-sensitive
ecoli_cnn	CNN/Mega	E. coli six-drug panel	23	6	4	18	8	9
ecoli_lgbm_multi	LightGBM multi-task	E. coli six-drug panel	23	6	4	18	7	10
ecoli_lgbm_single	LightGBM single-task	E. coli six-drug panel	23	6	4	18	6	11
weis_lr_ecoli_temporal_a2018	Weis LR	E. coli six-drug temporal A-2018 panel	6	6	1	4	2	2
saureus_cnn	CNN/Mega	S. aureus oxacillin panel	25	7	4	23	12	7
saureus_lgbm_multi	LightGBM multi-task	S. aureus oxacillin panel	25	7	4	23	13	7
saureus_lgbm_single	LightGBM single-task	S. aureus oxacillin panel	25	7	4	23	14	8
klebsiella_cnn	CNN/Mega	K. pneumoniae six-drug panel	24	6	4	15	4	11
weis_lr_klebsiella_ceftriaxone	Weis LR	K. pneumoniae ceftriaxone official-panel extension	3	1	3	3	2	0
weis_lr_ecoli	Weis LR	E. coli Weis-style panel	17	6	3	9	4	5

The atlas combines background-matched audit rows, exact-background baselines, calibration checks and sensitivity summaries. Retained/partial counts include rows classified as retained or partial residual signal by the atlas rules.

Table 14: Representative organism-extension audit rows.

Organism/drug	Model	Site	Raw AUC	Centered AUC	Background AUC	Retention (%)	Strata	Interpretation
S. aureus / Oxacillin	CNN/Mega	A-2018	0.838	0.763	0.762	60.1	9	strong within background signal
S. aureus / Oxacillin	CNN/Mega	DRIAMS-B	0.775	0.864	0.795	52.6	1	background only competitive but residual signal
S. aureus / Oxacillin	CNN/Mega	DRIAMS-C	0.725	0.699	0.627	59.6	2	moderate within background signal
S. aureus / Oxacillin	Weis LR	DRIAMS-B	0.667	0.667	–	100.0	1	caution low matched support
S. aureus / Oxacillin	Weis LR	DRIAMS-C	0.543	0.543	–	100.0	1	weak raw signal
K. pneumoniae / Ceftriaxone	CNN/Mega	A-2018	0.804	0.500	0.981	3.3	1	background only competitive and weak residual
K. pneumoniae / Ceftriaxone	CNN/Mega	DRIAMS-B	0.404	–	1.000	0.0	0	not interpretable or sparse
K. pneumoniae / Ceftriaxone	CNN/Mega	DRIAMS-C	0.698	–	1.000	0.0	0	not interpretable or sparse
K. pneumoniae / Ceftriaxone	CNN/Mega	DRIAMS-D	0.630	0.424	0.973	81.8	2	background only competitive and weak residual
K. pneumoniae / Ceftriaxone	Weis LR	DRIAMS-B	0.509	0.509	–	100.0	1	near chance or ambiguous
K. pneumoniae / Ceftriaxone	Weis LR	DRIAMS-C	0.672	0.672	–	100.0	1	cautionary moderate within background signal
K. pneumoniae / Ceftriaxone	Weis LR	DRIAMS-D	0.596	0.596	–	100.0	1	weak or partial within background signal

Representative rows test whether the audit is only an E. coli/ciprofloxacin story. S. aureus/oxacillin is a second-organism retained-signal case; K. pneumoniae/ceftriaxone is a third-organism background-sensitive or sparse-control case. Weis LR rows are compatibility exports; background-only AUC was not computed for those rows.

Table 15: Source-data files distributed with the manuscript package.

Display item	File	Description
Figure 2	`source_data_fig2_primary_background_audit.csv`	Primary raw and background-centered AU
Figure 3	`source_data_fig3_model_family_replication.csv`	CNN, LGBM and Weis/Borgwardt model-
Figure 4	`source_data_fig5a_public_wgs_maldi_auc.csv`	Public WGS-linked MALDI AUC values.
Figure 4	`source_data_fig5b_proteomic_biomarker_enrichment.csv`	ST131 biomarker enrichment values.
Supp. Fig. S1	`source_data_fig4_cross_resistance_edges_all_sites.csv`	Cross-resistance edges used for the phi hea
Supp. Fig. S4	`source_data_fig6_three_way_decomposition.csv`	Raw, co-resistance-only and background-c